DATASET 2: UNIVERSITY ENROLMENT RATE (Regression)

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SETTING A WORKING DIRECTORY

```
setwd("C:/Users/ADMIN/OneDrive/Desktop/caleb makini")
```

load libraries

```
library(tidyverse)
library(dplyr)
library(ggplot2)
library(ggthemes)
library(psych)
library(sjmisc)
library(kableExtra)
library(knitr)
library(tidyr)
library(reshape2)
library(formatR)
library(corrplot)
```

Section A:

load dataset

Q1: Report number of rows

```
data <- read.csv("Enrollment Rate.csv")
cat("Number of rows:", nrow(data), "\n")
```

```
Number of rows: 326
```

```
cat("Number of columns:", ncol(data), "\n")
```

```
Number of columns: 6
```

Q2: check structure

```
str(data)
```

```
'data.frame':   326 obs. of  6 variables:
 $ Year              : int  1700 1701 1702 1703 1704 1705 1706 1707 17
08 1709 ...
 $ GDP_per_Capita    : int  12000 12800 13500 14000 13200 13800 14600
15200 15900 16500 ...
 $ Education_Spending: num  3.2 3.4 3.6 3.8 3.9 4.1 4.3 4.5 4.7 4.9
...
 $ Unemployment_Rate : num  8.5 8.1 7.8 7.4 9.2 9 8.6 8.3 8 7.6 ...
 $ Population_Young  : num  22.1 21.8 21.5 21.1 20.9 20.6 20.2 19.9 1
9.5 19.2 ...
 $ Enrolment_Rate    : num  28.4 29.6 31.2 32.8 31.5 33.7 35.4 37.1 3
8.9 40.6 ...
```

```
data$GDP_per_Capita <- as.numeric(data$GDP_per_Capita)
cat("\n Report: All columns should be numeric/integer. Year is intege
r, others numeric.")
```

```

 Report: All columns should be numeric/integer. Year is integer, other
s numeric.
```

Q3: Check and remove for and duplicate

```
dup_idx <- duplicated(data)
if (sum(dup_idx) == 0) {
    cat("No duplicates in any numerical variable.\n")
} else {
    cat("Number of duplicated rows:", sum(dup_idx), "\n")
    print(data[dup_idx, , drop = FALSE])
}
```

```
No duplicates in any numerical variable.
```

Q4: use iqr method to identify outliers

detection using 1.5 IQR

```
detect_outliers_IQR <- function(x) {
    Q1 <- quantile(x, 0.25, na.rm = TRUE)
    Q3 <- quantile(x, 0.75, na.rm = TRUE)
    IQR_val <- Q3 - Q1
    lower <- Q1 - 1.5 * IQR_val
    upper <- Q3 + 1.5 * IQR_val
    which(x < lower | x > upper)
}
outlier_idx <- detect_outliers_IQR(data$GDP_per_Capita)
outliers_gdp <- data$GDP_per_Capita[outlier_idx]
outs <- outlier_idx[outlier_idx > 0]
if (length(outs) == 0) {
    cat("No outliers detected in any numerical variable.\n8i      Sinc
e no outliers, We Keep all values.")
} else {
    cat("Outliers detected:\n")
    print(outs)
}
```

```
No outliers detected in any numerical variable.
8i      Since no outliers, We Keep all values.
```

SECTION B: Missing-Value Treatment

Q5. Count missing values per column and

impute the missing values.

check for missiing values

```
mval_per_col <- colSums(is.na(data))
misss <- mval_per_col[mval_per_col > 0]
if (length(misss) == 0) {
    cat("No missing values detected in any numerical variable.\n3a
Hence no imputation of the missing values.")
} else {
    cat("Missing values detected:\n")
    print(misss)
}
```

```
No missing values detected in any numerical variable.
3a      Hence no imputation of the missing values.
```

Q6: Verification that no missing values exists

```
if (length(misss) == 0) {
    cat("No missing values detected in any numerical variable.\n0f
Hence Verified.")
} else {
    cat("Missing values detected:\n")
    print(misss)
}
```

```
No missing values detected in any numerical variable.
0f      Hence Verified.
```

SECTION C: Descriptive Statistics

Q7. Compute mean, median, standard deviation, min, max, and IQR for all

numerical variables.

```
num_vars <- c("GDP_per_Capita", "Education_Spending", "Unemployment_Ra
te", "Enrolment_Rate",
    "Population_Young")
num_vars <- data %>%
    select(where(is.numeric))
desc_stats <- num_vars %>%
    summarise(across(everything(), list(mean = ~mean(.x, na.rm = TRU
E), median = ~median(.x,
        na.rm = TRUE), sd = ~sd(.x, na.rm = TRUE), min = ~min(.x, na.r
m = TRUE),
        max = ~max(.x, na.rm = TRUE), IQR = ~IQR(.x, na.rm = TRUE))))
%>%
    pivot_longer(everything(), names_to = c("variable", "stat"), names
_sep = "_(?=[^_]+$)") %>%
    pivot_wider(names_from = stat, values_from = value) %>%
    arrange(variable)
desc_stats
```

```
# A tibble: 6 × 7
  variable               mean  median      sd     min     max     IQR
  <chr>                 <dbl>   <dbl>   <dbl>   <dbl>   <dbl>   <dbl>
1 Education_Spending     7.93     8.4    2.38     3.2    11.4     4.1
2 Enrolment_Rate        59.4     62.8   15.2     28.4    80      25.6
3 GDP_per_Capita     24731.   25300   6909.   12000   35800   11900
4 Population_Young      14.9     14.3    3.47     9.8    22.1     5.5
5 Unemployment_Rate      5.91     5.7    1.69     3.5     9.2     3.1
6 Year                1862.    1862.    94.3   1700    2025     162.
```

Q8. Compute skewness and kurtosis for each variable. Interpret the results.

```
library(e1071)
skew_kurt <- num_vars %>%
    summarise(across(everything(), list(skewness = ~skewness(.x, na.rm
= TRUE), kurtosis = ~kurtosis(.x,
        na.rm = TRUE)))) %>%
    pivot_longer(everything(), names_to = c("variable", "stat"), names
_sep = "_(?=[^_]+$)") %>%
    pivot_wider(names_from = stat, values_from = value) %>%
    arrange(variable)
skew_kurt
```

```
# A tibble: 6 × 3
  variable           skewness kurtosis
  <chr>                 <dbl>    <dbl>
1 Education_Spending   -0.379   -1.06
2 Enrolment_Rate       -0.445   -1.02
3 GDP_per_Capita       -0.128   -1.18
4 Population_Young      0.409   -0.927
5 Unemployment_Rate     0.265   -1.21
6 Year                  0       -1.21
```

- All six variables are roughly symmetric (skewness close to 0) and have negative kurtosis (around −1), meaning flat, thin-tailed distributions with no serious outliers. So the data are well-behaved: no transformation is needed, Pearson correlation is appropriate, and standard models can be used directly.

Q9:What is the range of `Enrolment_Rate`? Is there sufficient variance for regression modeling?

```
target <- "Enrolment_Rate"
predictors <- c("GDP_per_Capita", "Education_Spending", "Unemployment_
Rate", "Population_Young")
cat("Range:", range(data[[target]]), "\n")
```

```
Range: 28.4 80
```

- Enrolment_Rate ranges from 28.4 to 80.0 (≈52-point spread), providing ample variance for regression.
- no boundary compression, no skew distortion
- so it is suitable as a target or predictor, provided sample size and linearity are also adequate.